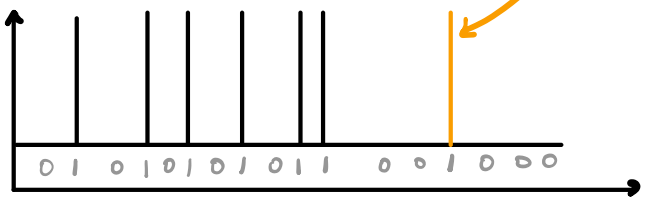
Chapter 3 : Resting Potential 静息电位

firing ≈ nerve impulse ≈ action potential (AP) ≈ spike

动作电位



electrical conductance (g) 电导.

ability of movement/migrate of particles.

neuron: permeability of membrane.



——→ determined by functional ion channel.

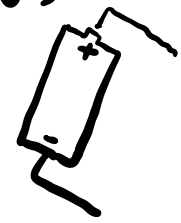
electrical resistance (R) $R = 1/g$

inability of movement/migration of particles.

electrical potential ≈ voltage difference. (V)

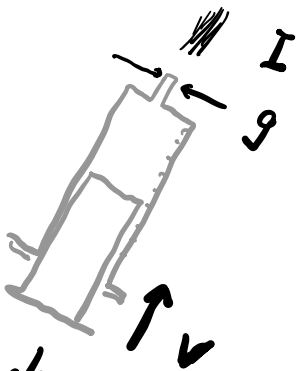
force on charged particle

determined by anode (+) and cathode (-)



Ohm's Law: $I = g \times V$ → force

electrical current ↙ ↘ conductance



Steps on Resting Potential Research.

I. Concentration Gradient across membrane.

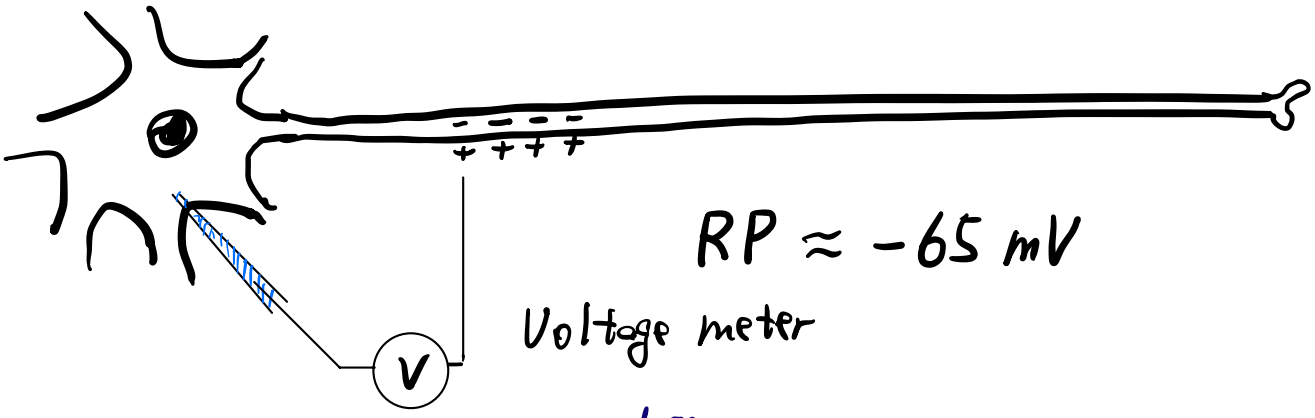
high K^+ inside, high Na^+ outside

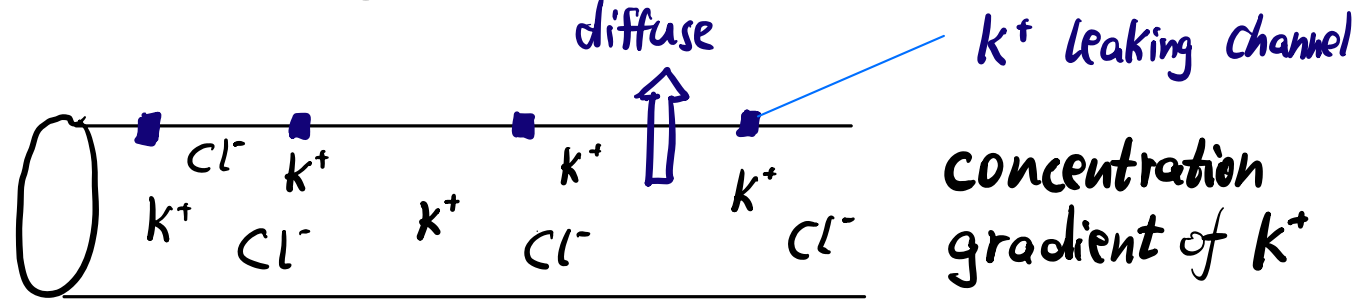
Ⅱ. Value of Resting Potential

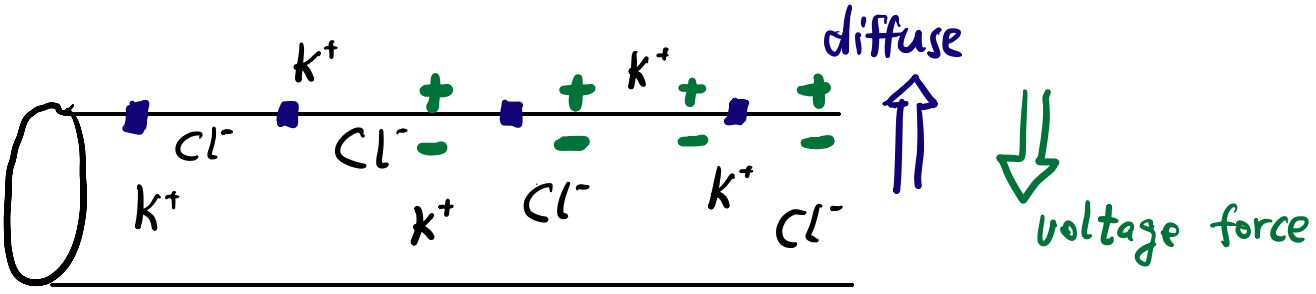
内负外正

$\frac{----}{++++}$ inside / outside

Ⅲ $Na^+ - K^+ - ATPase \approx Na^+ - K^+ - Pump$







0 mM $[K^+]_o$

↑ 0 mV

100 mM $[K^+]_i$

↑ 0.00001 mM $[K^+]_o$ ↓ -80mV

99.99999 mM $[K^+]_i$

外流

- tiny amount of K^+ efflux leads to great voltage difference.
- given the initial concentration gradient, the **equilibrium potential** can be calculated.

Nernst Equation:

$$E_{ion} = \frac{RT}{zF} \log_{10} \frac{[ion]_o}{[ion]_i}$$

$[ion]_o$ → outside, $[ion]_i$ → inside

$\frac{RT}{zF} = 61.54$

$[K^+]_i \approx 100\,mM \quad [K^+]_o \approx 5\,mM \quad \frac{1}{20}$

$$E_{K^+} = 61.54 \log_{10}\left[\frac{5}{100}\right]$$

$$= 61.54 \times (-1.3) \approx -80\,mV$$

$[Na^+]_i \approx 15\,mM. \quad [Na^+]_o \approx 150\,mM \quad \frac{10}{1}$

$$E_{Na^+} = 61.54 \log_{10}\left[\frac{150}{15}\right] = 61.54\,mV$$

Resting Potential (RP) $\in (-80\,mV,\ +61.54\,mV)$

$-65\,mV$ $\qquad E_{K^+} \qquad E_{Na^+}$

Goldman Equation

$$V_m = \log_{10}\left[\frac{P_{K^+}\cdot[K^+]_o + P_{Na^+}[Na^+]_o}{P_{K^+}[K^+]_i + P_{Na^+}[Na^+]_i}\right] \cdot \frac{RT}{zF}$$

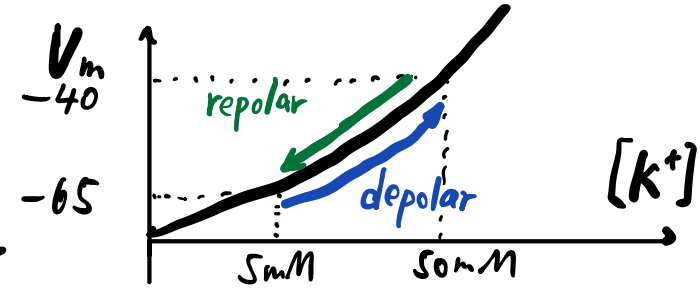
P → Permeability. active ion channel.

$P_{K^+} = P_{Na^+} \times 40$ during resting potential.

$$V_m = 61.54 \cdot \log_{10} \frac{40\times5 + 1\times150}{40\times100 + 1\times15} = 61.54 \times \log_{10} \frac{350}{4015}$$

V_m ↓ membrane potential

$= -65\,mV.$



polar

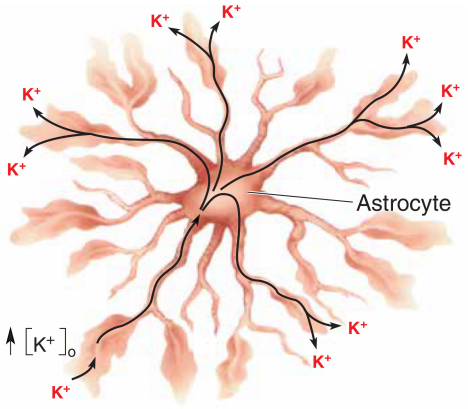
$[K^+]_o \approx 5mM$, under regulation of:

- **Blood-Brain-Barrier**
  regulate transport of K^+ from blood to CSF (cerebrospinal fluid)

- **astrocyte act as spatial-buffer** for K^+

- Intravenous injection of KCl (IV) is lethal.

"Myogenic Tissue" ⟶ "heart failure"

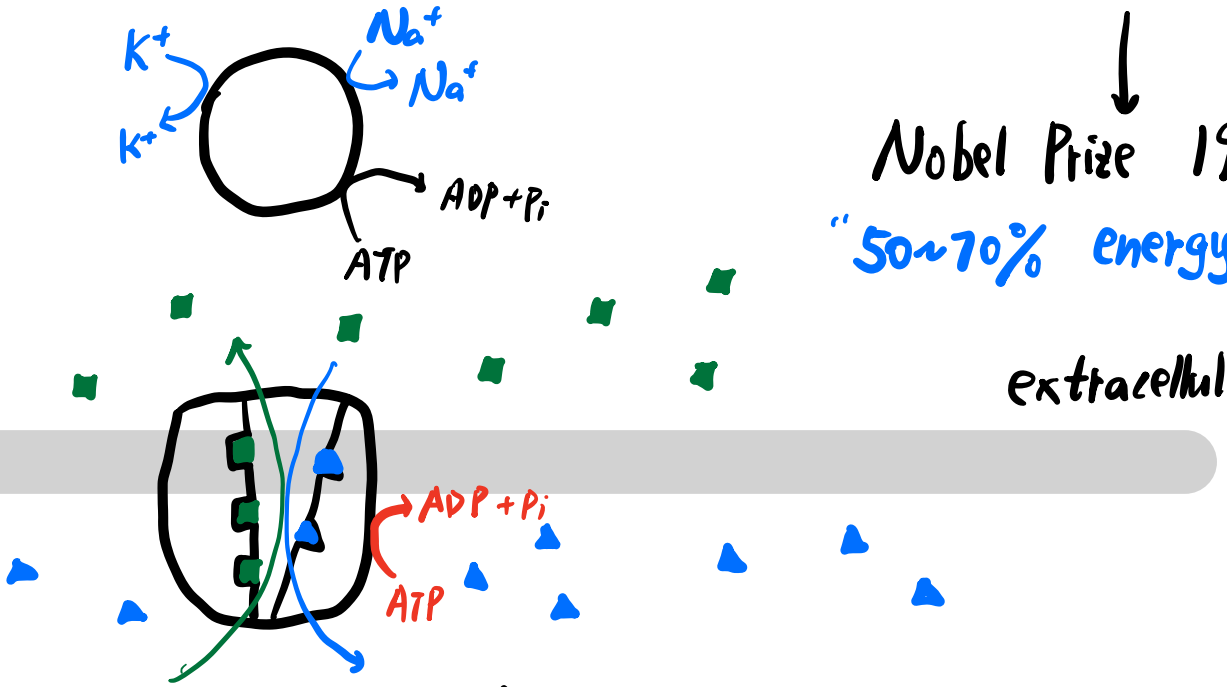


Na^+-K^+-Pump. Na^+-K^+-ATPase. Skou, 1957

↓

Nobel Prize 1997

"50~70% energy consumed"



3.2.1 $3Na^+$ efflux, $2K^+$ influx, 1ATP hydrolyzed

	intracellular	extracellular	O:I ratio	E_{ion}
Na^+	15mM	150mM	10:1	+61.54 mV
K^+	100mM	5mM	1:20	−80 mV

Driven force = $V_m - E_{ion}$

K^+ $\quad -65 - (-80) \Longrightarrow +15mV$

Na^+ $\quad -65 - (+61.54) \Longrightarrow -126.54mV$

$I = g \times V$

driven force